GATE ME 90 Days Challenge

Day 28/90 · Practice Set with Detailed Solutions

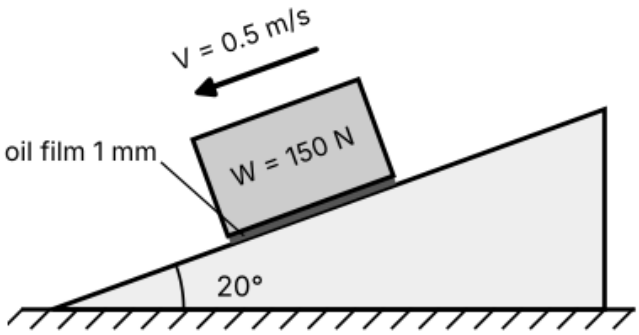
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Fluid Mechanics	Fluid Properties & Statics	79 days to GATE

How to use this set: 15 questions (5 MCQ, 10 NAT) on viscosity, surface tension and compressibility, hydrostatic pressure and manometry, forces on plane and curved submerged surfaces, buoyancy and metacentric height, and liquids in rigid-body rotation. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

A block weighing 150 N with a 0.4 m × 0.4 m base slides down a plane inclined at 20° on an oil film 1 mm thick, as shown. Its steady (terminal) speed is 0.5 m/s. Assuming a linear velocity profile in the film, the dynamic viscosity of the oil (in Pa·s, round off to two decimal places) is _____



Q2 NAT

A clean glass tube of 2 mm internal diameter is dipped vertically in water (σ = 0.073 N/m, contact angle 0°). The capillary rise (in mm, round off to one decimal place) is _____

Q3 MCQ

The excess pressure inside a soap bubble of diameter d (surface tension σ) over the outside pressure is

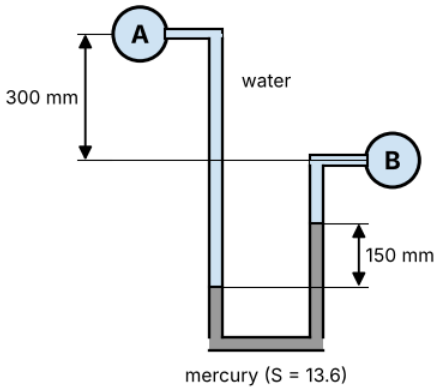
(A) 4σ/d
(B) 8σ/d
(C) 2σ/d
(D) 16σ/d

Q4 NAT

The bulk modulus of water is 2.2 GPa. The increase in pressure needed to reduce a given volume of water by 0.5 % (in MPa) is _____

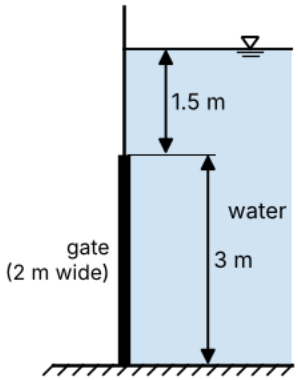
Q5 NAT

Pipes A and B carry water and are connected by a differential U-tube mercury manometer, as shown. The centre of A is 300 mm above the centre of B and the mercury level in the limb on B's side stands 150 mm higher than in the limb on A's side. The pressure difference $p_A - p_B$ (in kPa, round off to two decimal places) is _____



Q6 NAT

A vertical rectangular gate 2 m wide and 3 m high holds water on one side; its top edge is 1.5 m below the free surface, as shown. The depth of the centre of pressure below the free surface (in m, round off to two decimal places) is _____

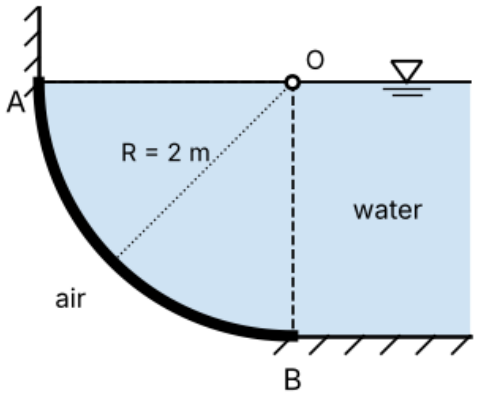


Q7 NAT

A circular plate of 1.2 m diameter is immersed in water with its plane inclined at 60° to the horizontal and its centre 2 m below the free surface. The vertical depth of its centre of pressure below the free surface (in m, round off to three decimal places) is _____

Q8 NAT

The quarter-circle gate AB shown (radius 2 m, width 3 m perpendicular to the paper) holds water above it, with its centre O at the free surface. The magnitude of the resultant hydrostatic force on the gate (in kN, round off to one decimal place) is _____



Q9 MCQ

For a plane surface submerged vertically (or inclined) in a liquid, the centre of pressure lies below the centroid because

(A) the liquid surface is free
(B) the plate is thin
(C) the pressure increases with depth
(D) the area of the plate is large

Q10 NAT

A rectangular wooden pontoon 1 m long, 0.5 m wide and 0.4 m high (specific gravity 0.7) floats in water with its 0.4 m side vertical. Its metacentric height for rolling about the longitudinal axis (in mm, round off to one decimal place) is _____

Q11 MCQ

Which statement about the stability of bodies in a liquid is correct?

(A) A floating body is stable if its metacentre lies above its centre of gravity
(B) A submerged body is stable if its centre of gravity lies above its centre of buoyancy
(C) A floating body is stable only if its centre of gravity lies below its centre of buoyancy
(D) A floating body is stable if its metacentre lies below its centre of gravity

Q12 NAT

An open tank contains a 2 m layer of oil (specific gravity 0.85) floating on 3 m of water. The gauge pressure at the bottom of the tank (in kPa, round off to two decimal places) is _____

Q13 NAT

An open cylindrical tank of 0.4 m diameter, partly filled with water, is rotated about its vertical axis at 150 rpm. With no water spilling, the difference between the water levels at the wall and at the centre (in m, round off to two decimal places) is _____

Q14 MCQ

A stone weighs 50 N in air, 30 N when fully immersed in water and 34 N when fully immersed in an oil. The specific gravity of the oil is

(A) 0.8
(B) 0.68
(C) 1.25
(D) 0.6

Q15 MCQ

A fluid whose apparent viscosity decreases as the rate of shear increases (for example, blood or paint) is called

(A) dilatant
(B) pseudoplastic (shear-thinning)
(C) Bingham plastic
(D) Newtonian

Answer Key

Q	Type	Answer
Q1	NAT	0.64 (accept 0.63 to 0.65)
Q2	NAT	14.9 (accept 14.7 to 15.1)
Q3	MCQ	(B) 8σ/d
Q4	NAT	11
Q5	NAT	15.60 (accept 15.50 to 15.70)
Q6	NAT	3.25 (accept 3.23 to 3.27)
Q7	NAT	2.034 (accept 2.032 to 2.036)
Q8	NAT	109.6 (accept 109.0 to 110.2)
Q9	MCQ	(C) the pressure increases with depth
Q10	NAT	14.4 (accept 14.0 to 14.8)
Q11	MCQ	(A) A floating body is stable if its metacentre lies above its centre of gravity
Q12	NAT	46.11 (accept 45.90 to 46.30)
Q13	NAT	0.50 (accept 0.49 to 0.51)
Q14	MCQ	(A) 0.8
Q15	MCQ	(B) pseudoplastic (shear-thinning)

Detailed Solutions

Q1 **Answer: 0.64 (accept 0.63 to 0.65)**

Concept: At terminal speed the weight component along the plane equals the viscous drag: $W \sin\theta = \tau A = \mu(V/h)A$.

$W \sin\theta = 150 \times \sin 20° =$ **51.30 N**

$\mu = W \sin\theta \times h/(A\,V) = 51.30 \times 0.001/(0.16 \times 0.5) =$ **0.64 Pa·s**

Exam tip: 1 Pa·s = 10 poise, so this oil has $\mu = 6.4$ P.

Q2 **Answer: 14.9 (accept 14.7 to 15.1)**

Concept: Capillary rise $h = 4\sigma\cos\theta/(\rho g d)$, from balancing surface tension round the perimeter against the weight of the raised column.

$h = 4 \times 0.073 \times 1/(1000 \times 9.81 \times 0.002) =$ **0.01488 m = 14.9 mm**

Exam tip: Mercury ($\theta \approx 130°$) gives a negative h: capillary depression. Halving the diameter doubles the rise.

Q3 **Answer: (B) 8σ/d**

Concept: A soap film has **two** surfaces (inside and outside). Force balance on half the bubble: $\Delta p \times \pi d^2/4 = 2 \times \sigma\pi d$.

$\Delta p =$ **8σ/d**

Exam tip: Liquid droplet (one surface): 4σ/d; liquid jet (cylinder): 2σ/d.

Q4 **Answer: 11**

Concept: $K = -\Delta p/(\Delta V/V)$.

$\Delta p = K \times 0.005 = 2200 \text{ MPa} \times 0.005 =$ **11 MPa**

Exam tip: This is about 1100 m of water head for only 0.5 % volume change, which is why liquids are treated as incompressible.

Q5 **Answer: 15.60 (accept 15.50 to 15.70)**

Concept: Start at A, add ρgh going down, subtract going up, and end at B.

Take the lower mercury surface as datum and let B be y above it, so A is (y + 0.3) above it.

$p_A + \rho_w g(y + 0.3) - \rho_{Hg} g(0.15) - \rho_w g(y - 0.15) = p_B$

$p_A - p_B = 13600 \times 9.81 \times 0.15 - 1000 \times 9.81 \times 0.45 = 20012 - 4415 =$ **15.60 kPa**

Exam tip: The unknown y always cancels; if it does not, a sign is wrong.

Q6 **Answer: 3.25 (accept 3.23 to 3.27)**

Concept: $F = \rho g A \bar{h}$ and $h_{cp} = \bar{h} + I_G/(A\,\bar{h})$ for a vertical plane.

$\bar{h} = 1.5 + 1.5 = 3$ m, $A = 6$ m², $I_G = 2 \times 3^3/12 = 4.5$ m⁴

$h_{cp} = 3 + 4.5/(6 \times 3) =$ **3.25 m** (force $F = 9810 \times 6 \times 3 = 176.6$ kN)

Exam tip: The deeper the gate, the closer the centre of pressure moves to the centroid.

Q7 **Answer: 2.034 (accept 2.032 to 2.036)**

Concept: For an inclined plane: $h_{cp} = \bar{h} + I_G \sin^2\theta/(A\,\bar{h})$.

I_G/A for a circle $= d^2/16 = 1.44/16 =$ **0.09 m²**

$h_{cp} = 2 + 0.09 \times \sin^2 60°/2 = 2 + 0.03375 =$ **2.034 m**

Force: $F = \rho g A \bar{h} = 9810 \times 1.131 \times 2 = 22.19$ kN

Exam tip: The total force does not depend on the inclination; only the position of the centre of pressure does.

Q8 **Answer: 109.6 (accept 109.0 to 110.2)**

Concept: Horizontal component = force on the vertical projection; vertical component = weight of the water above the curved surface up to the free surface.

$F_H = \rho g\,(R/2)(R\,b) = 9810 \times 1 \times (2 \times 3) =$ **58.86 kN**

$F_V = \rho g\, b\, \pi R^2/4 = 9810 \times 3 \times \pi =$ **92.46 kN** (downward)

$F = \sqrt{(58.86^2 + 92.46^2)} =$ **109.6 kN**, at 57.5° to the horizontal, passing through O

Exam tip: On a circular surface every pressure force is normal to it, so the resultant always passes through the centre.

Q9 **Answer: (C) the pressure increases with depth**

Concept: The pressure distribution is triangular/trapezoidal, so the larger pressures on the lower part move the line of action of the resultant below the centroid by $I_G \sin^2\theta/(A\,\bar{h})$.

For a **horizontal** plane the pressure is uniform and the two points coincide.

Exam tip: The gap $I_G/(A\,\bar{h}) \to 0$ as the plate goes deeper.

Q10 **Answer: 14.4 (accept 14.0 to 14.8)**

Concept: GM = BM − BG with BM = I/V (I = second moment of the waterline plane about the roll axis).

Depth of immersion d = 0.7 × 0.4 = **0.28 m**; KB = 0.14 m, KG = 0.20 m

BM = $(L\, b^3/12)/(L\, b\, d) = b^2/(12d) = 0.25/3.36$ = **0.0744 m**

GM = 0.14 + 0.0744 − 0.20 = **0.0144 m = 14.4 mm** (stable)

Exam tip: Rolling uses the smaller dimension b; pitching about the transverse axis would use L^3 and be far more stable.

Q11 **Answer: (A) A floating body is stable if its metacentre lies above its centre of gravity**

Concept: For a floating body the shift of B on tilting creates a restoring couple if M lies above G (GM > 0).

For a fully **submerged** body B does not shift, so it is stable only if B is above G.

Exam tip: A floating body can be stable even with G above B (ships), thanks to the metacentre.

Q12 **Answer: 46.11 (accept 45.90 to 46.30)**

Concept: Pressures of stacked layers add: p = Σρgh.

p = 9810 × (0.85 × 2 + 1 × 3) = 9810 × 4.7 = **46.11 kPa**

Exam tip: Express every layer as an equivalent water head first: 0.85 × 2 = 1.7 m of water.

Q13 **Answer: 0.50 (accept 0.49 to 0.51)**

Concept: Forced vortex (rigid-body rotation): the free surface is a paraboloid with $z = \omega^2 r^2/(2g)$.

ω = 2π × 150/60 = **15.71 rad/s**

$\Delta z = 15.71^2 \times 0.2^2/(2 \times 9.81)$ = **0.503 m**

Exam tip: The volume of a paraboloid is half that of its circumscribing cylinder, so the level rises Δz/2 at the wall and falls Δz/2 at the centre.

Q14 **Answer: (A) 0.8**

Concept: Loss of weight = buoyancy = $\rho_{liquid}\, g\, V$.

In water: 50 − 30 = 20 N; in oil: 50 − 34 = 16 N

S_{oil} = 16/20 = **0.8** (and S_{stone} = 50/20 = 2.5)

Exam tip: This is the principle of the hydrometer and of finding the density of an irregular solid.

Q15 **Answer: (B) pseudoplastic (shear-thinning)**

Concept: Power-law fluid $\tau = K(du/dy)^n$: n < 1 pseudoplastic (shear-thinning), n > 1 dilatant (shear-thickening), n = 1 Newtonian.

A Bingham plastic (toothpaste) needs a yield stress before it flows.

Exam tip: Thixotropic fluids thin with the **time** of shearing, not just its rate.

What next?

Day 28 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 29/90: Fluid Mechanics, Fluid Kinematics & Bernoulli. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.